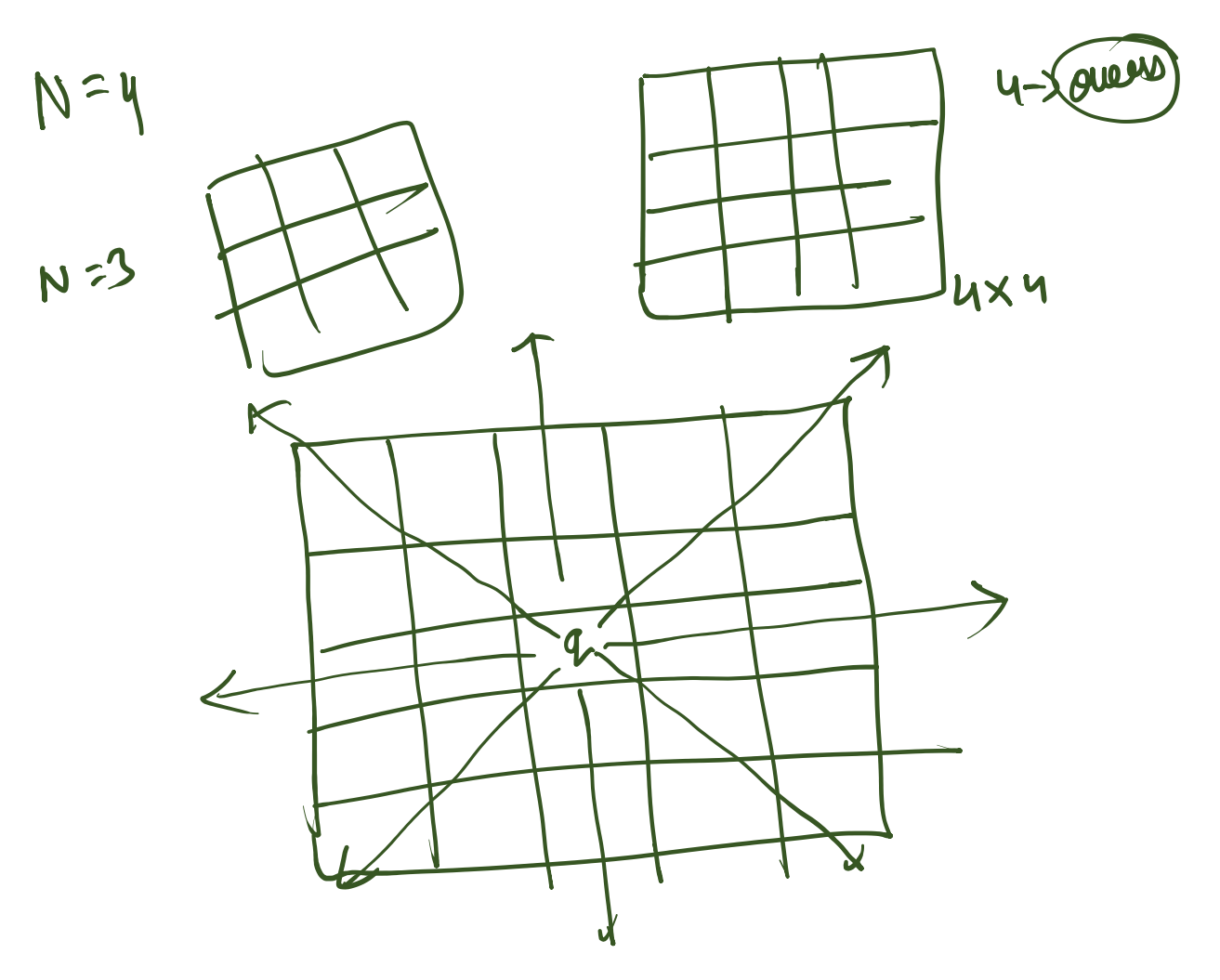
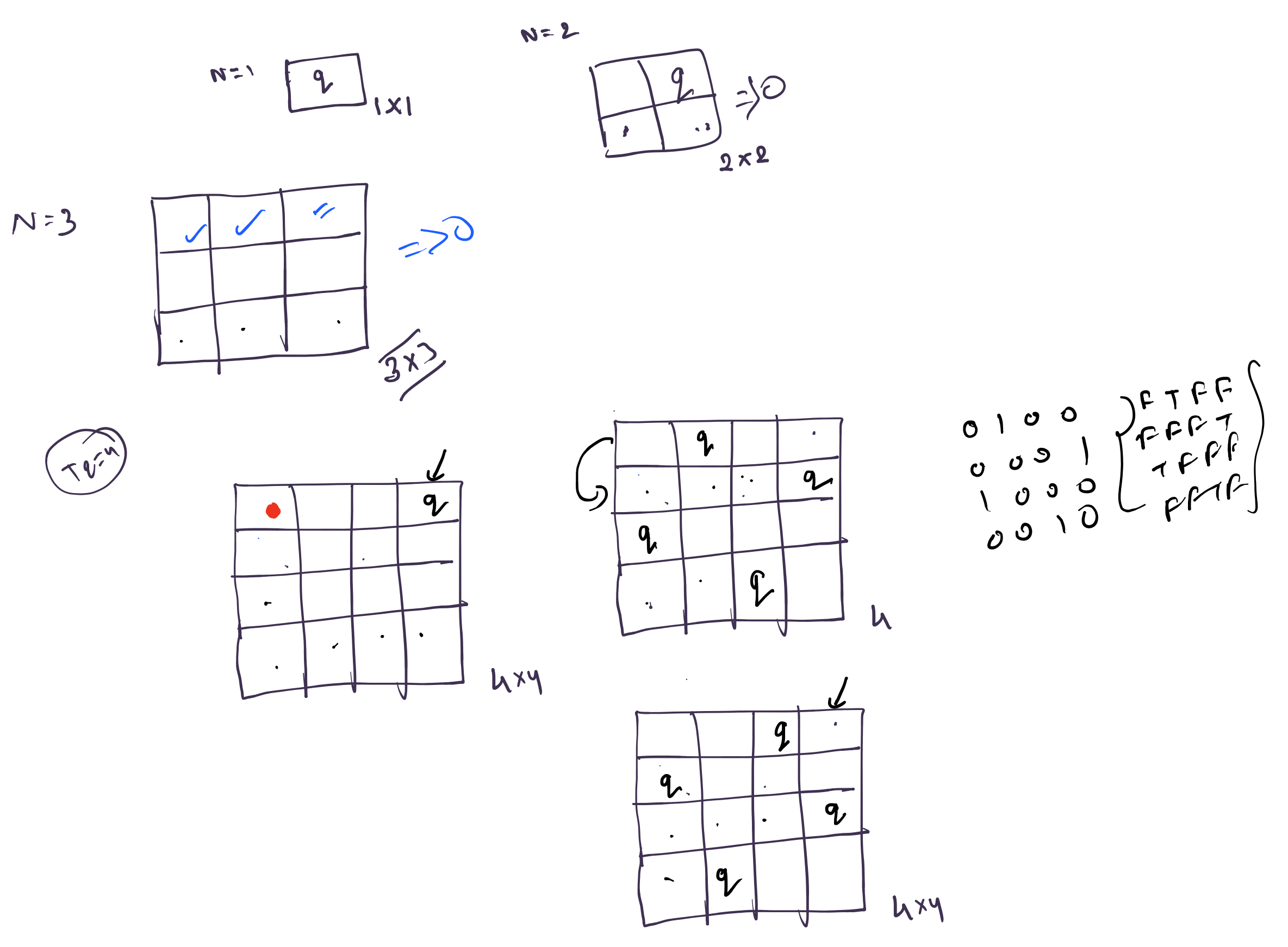
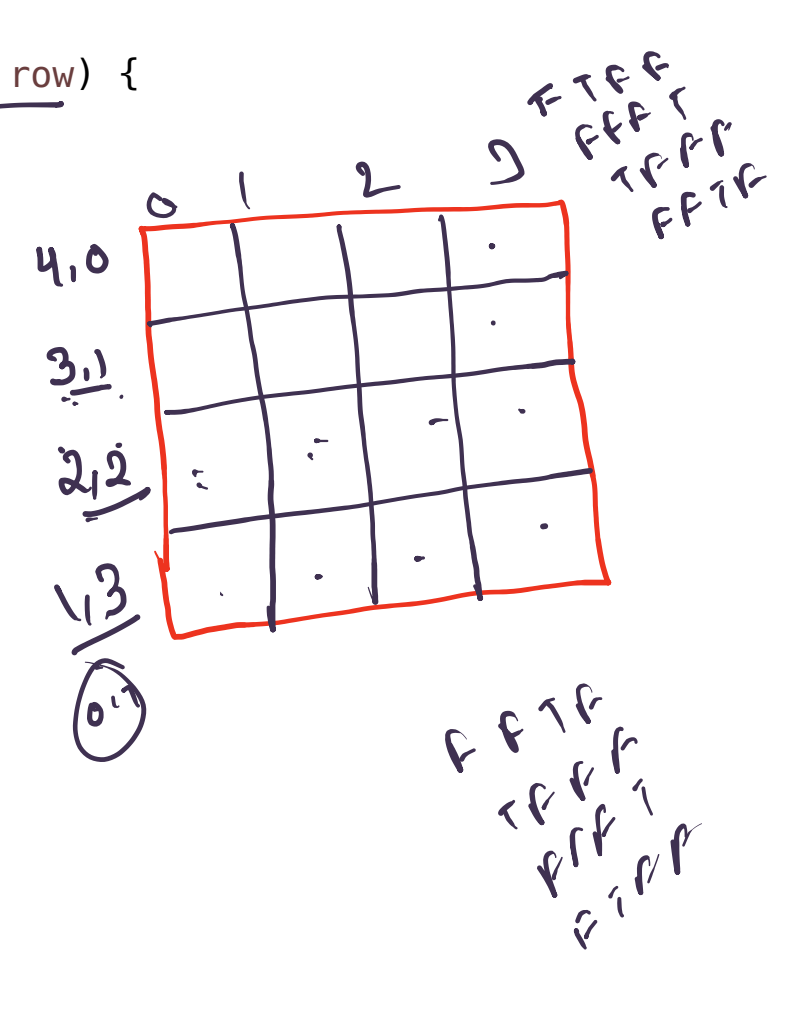
Lec-20

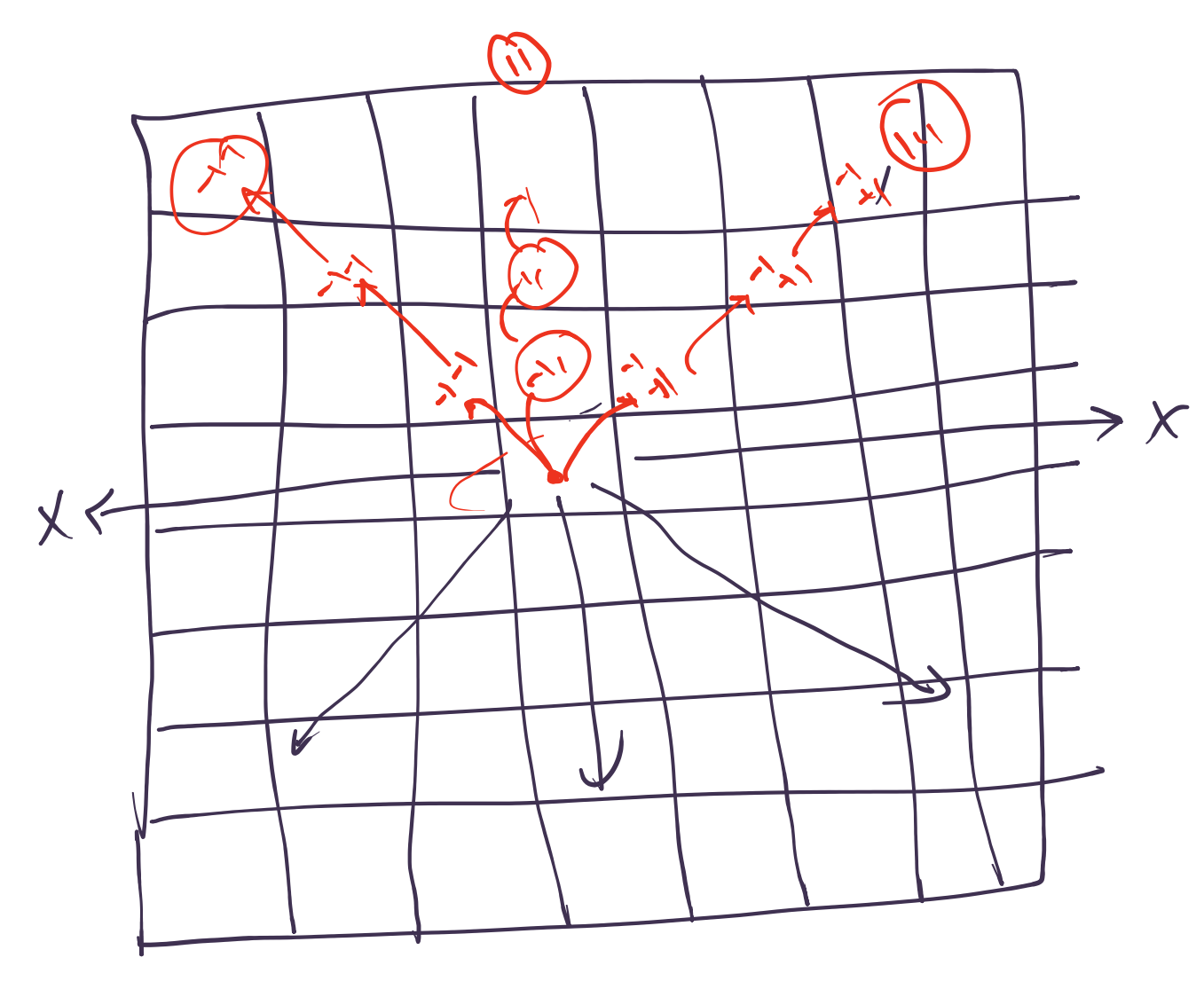
21/03/26 4:45 PM

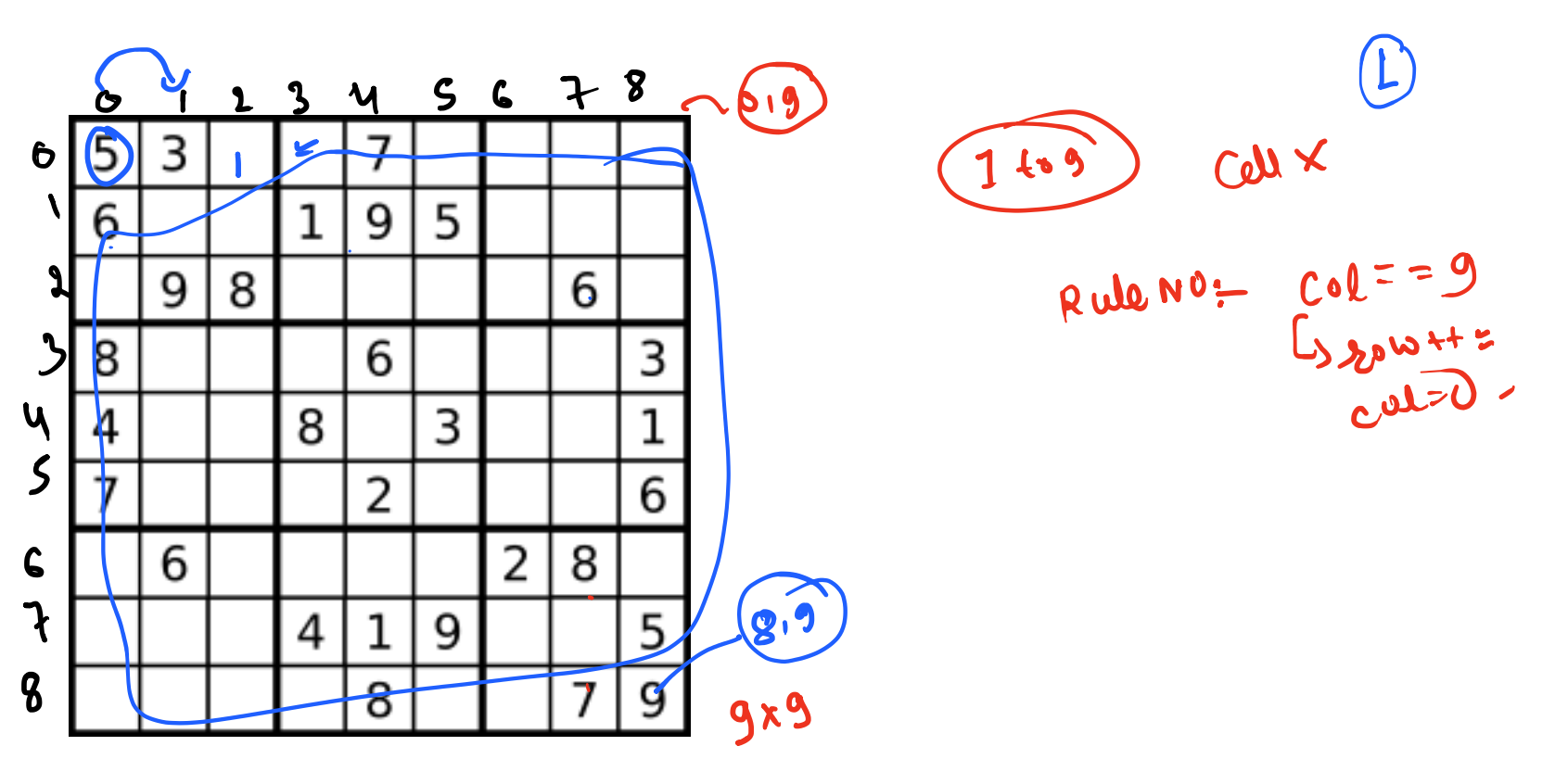

N=4
N=3
4→queens
4x4
q


N=1
1x1
N=2
=>0
2x2
N=3
=>0
3x3
Tc=4
4x4
4
4x4
0 1 0 0
0 0 0 1
1 0 0 0
0 0 1 0
FTFF
FFFT
TFFF
FFTF

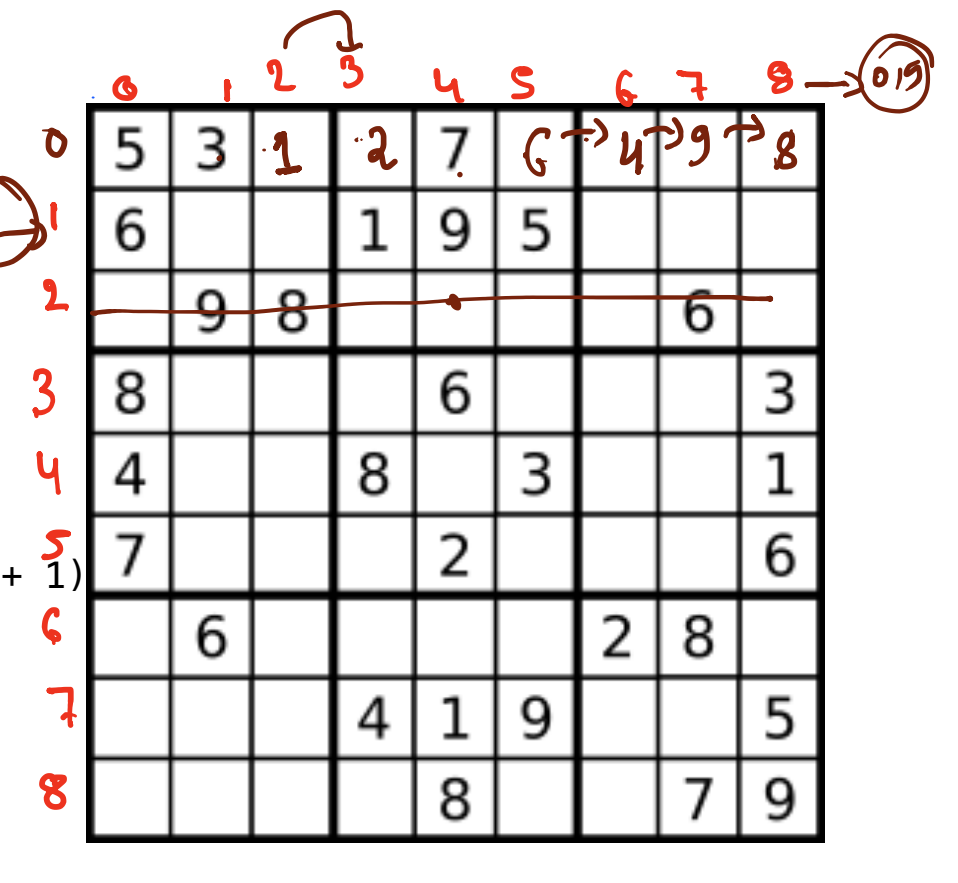
```
public static void Print(boolean[][] board, int tq, int row) {
    if (tq == 0) {
        Display(board);
        return;
    }
    for (int col = 0; col < board[0].length; col++) {
        if (isitsafe(board, row, col)) {
            board[row][col] = true;
            Print(board, tq - 1, row + 1);
            board[row][col] = false;
        }
    }
}
```

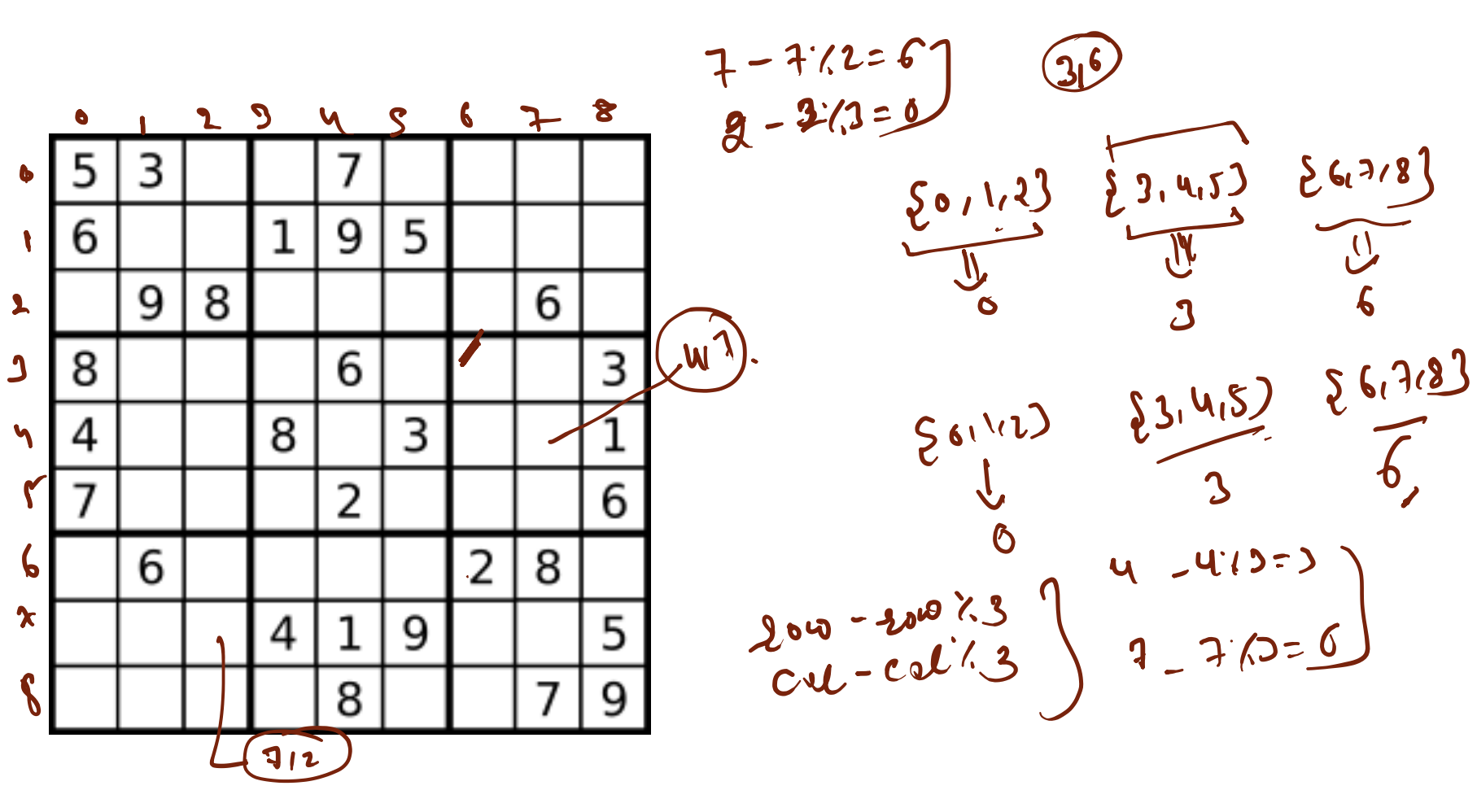

4,0
3,1
2,2
1,3
0,1
0 1 2 3
FTFF
FFFT
TFFF
FFTF
FFTF
TFFF
FFFT
FTFF

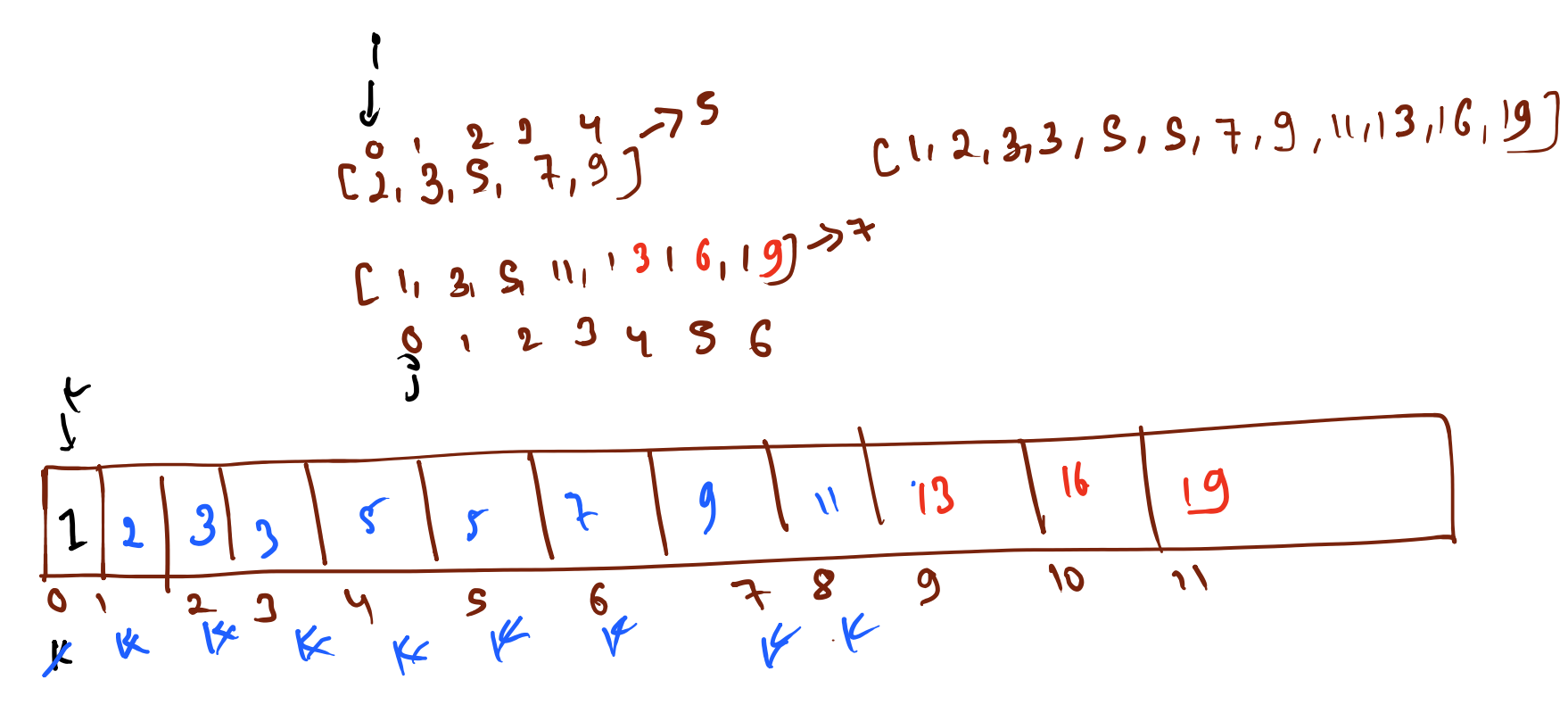



0 1 2 3 4 5 6 7 8
0,9
1 to 9
Cell X
Rule No:- col==9
row++
col=0
8,9
9x9

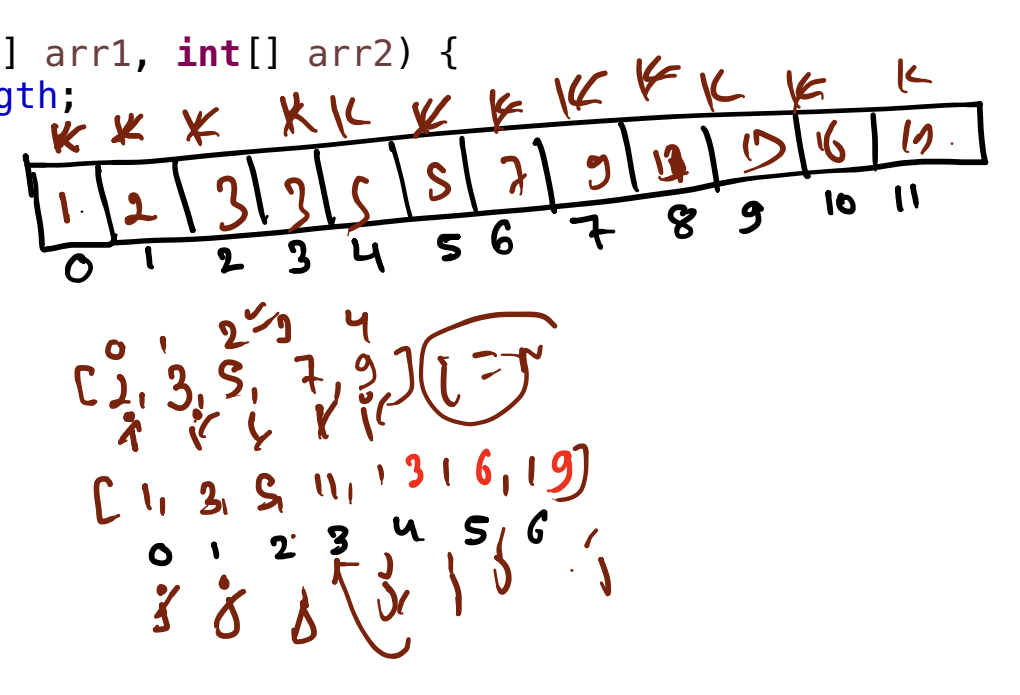
```
public static boolean Print(int[][] grid, int row, int col) {
    if (col == 9) {
        row++;
        col = 0;
    }
    if (row == 9) {
        return true;
    }
    if (grid[row][col] != 0) {
        return Print(grid, row, col + 1);
    } else {
        for (int val = 1; val <= 9; val++) {
            if (isitsafe(grid, row, col, val)) {
                grid[row][col] = val;
                boolean ans = Print(grid, row, col + 1);
                if (ans == true) {
                    return true;
                }

                grid[row][col] = 0;
            }
        }
    }
    return false;
}
```



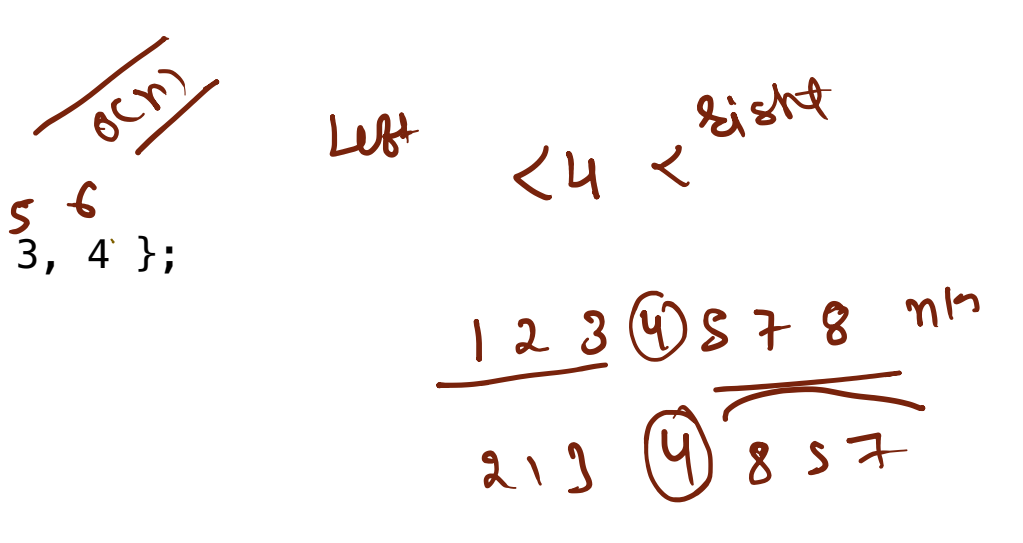

7 - 7%3 = 6
8 - 8%3 = 0
3,6
{0,1,2} 0
{3,4,5} 3
{6,7,8} 6
row - row%3
col - col%3
4 - 4%3 = 3
7 - 7%3 = 6
4,7
7,2

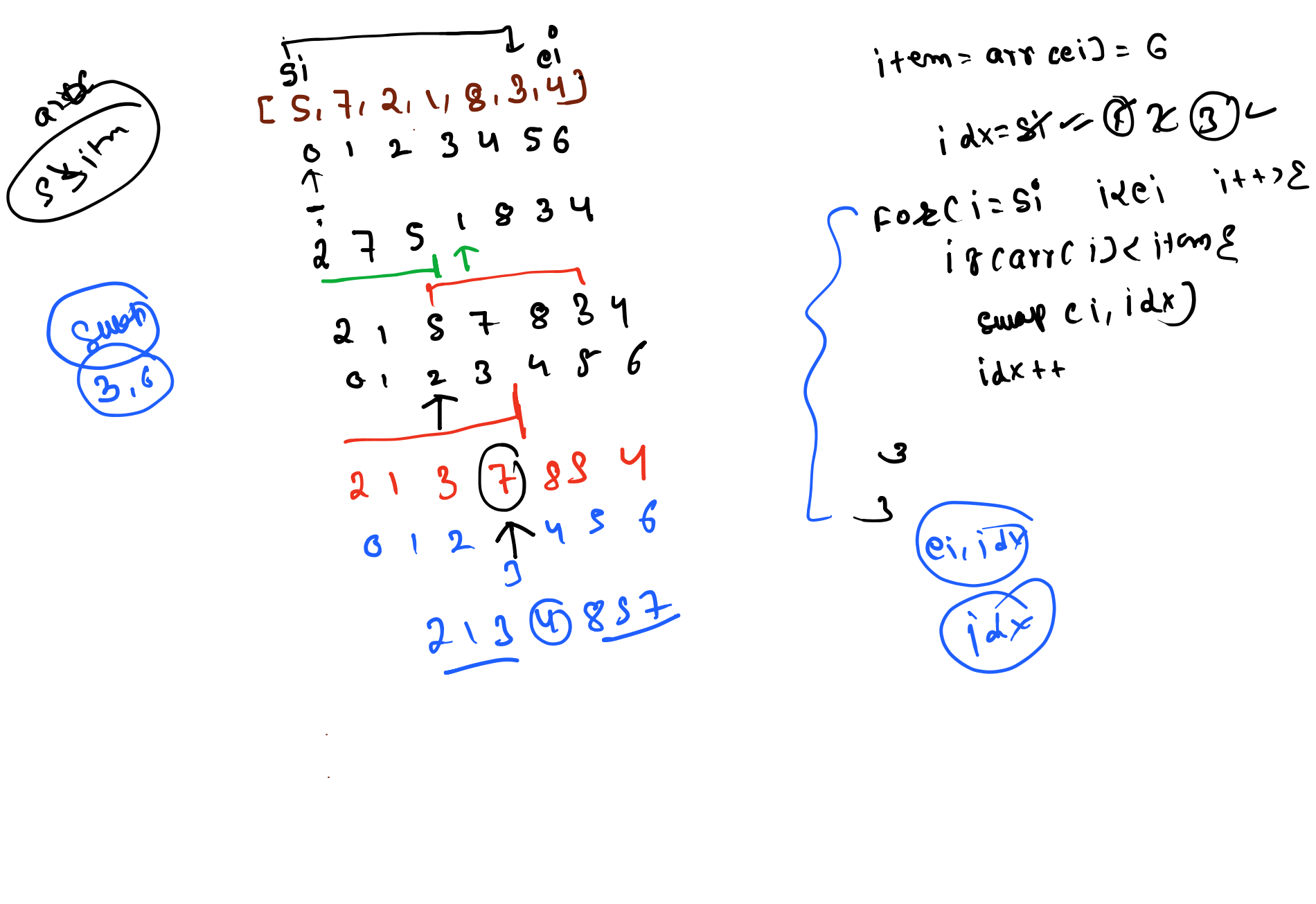

[2, 3, 5, 7, 9] →5
[1, 2, 3, 3, 5, 5, 7, 9, 11, 13, 16, 19]
[1, 3, 5, 11, 13, 16, 19] →7
1 2 3 3 5 5 7 9 11 13 16 19
0 1 2 3 4 5 6 7 8 9 10 11

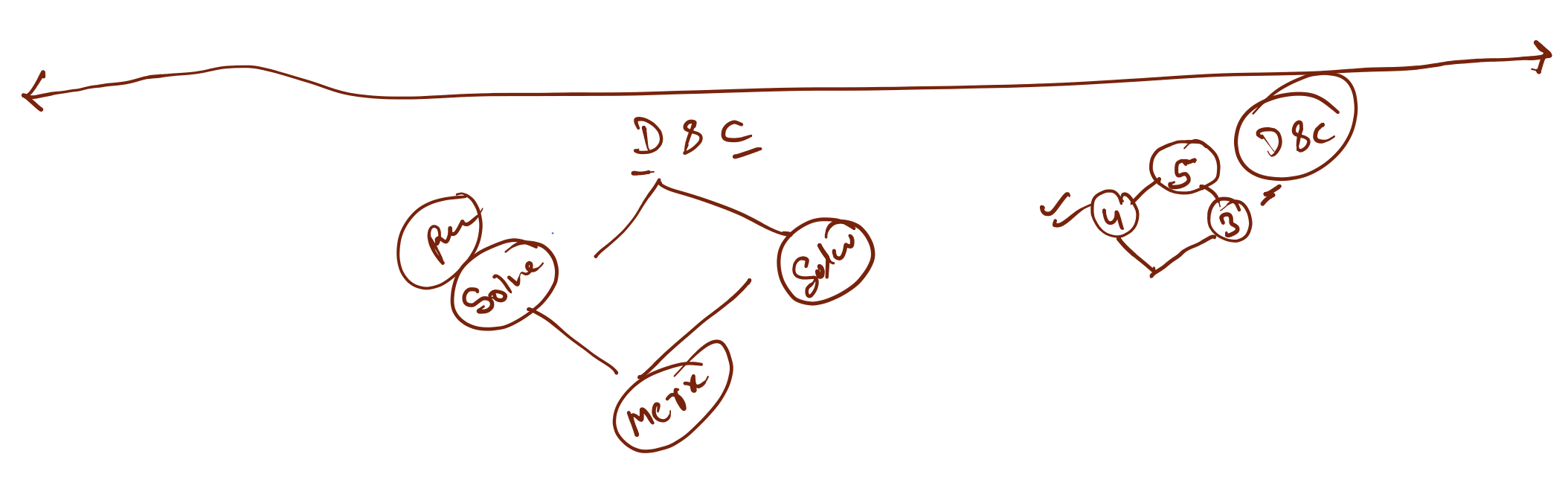
```
public static int[] Merge_Two_Array(int[] arr1, int[] arr2) {
    int n = arr1.length, m = arr2.length;
    int i = 0, j = 0, k = 0;
    int[] ans = new int[n + m];
    while (i < n && j < m) {
        if (arr1[i] < arr2[j]) {
            ans[k++] = arr1[i++];
        } else {
            ans[k++] = arr2[j++];
        }
    }
    while(i<n) {
        ans[k++] = arr1[i++];
    }
    while(j<m) {
        ans[k++] = arr2[j++];
    }
    return ans;
}
```

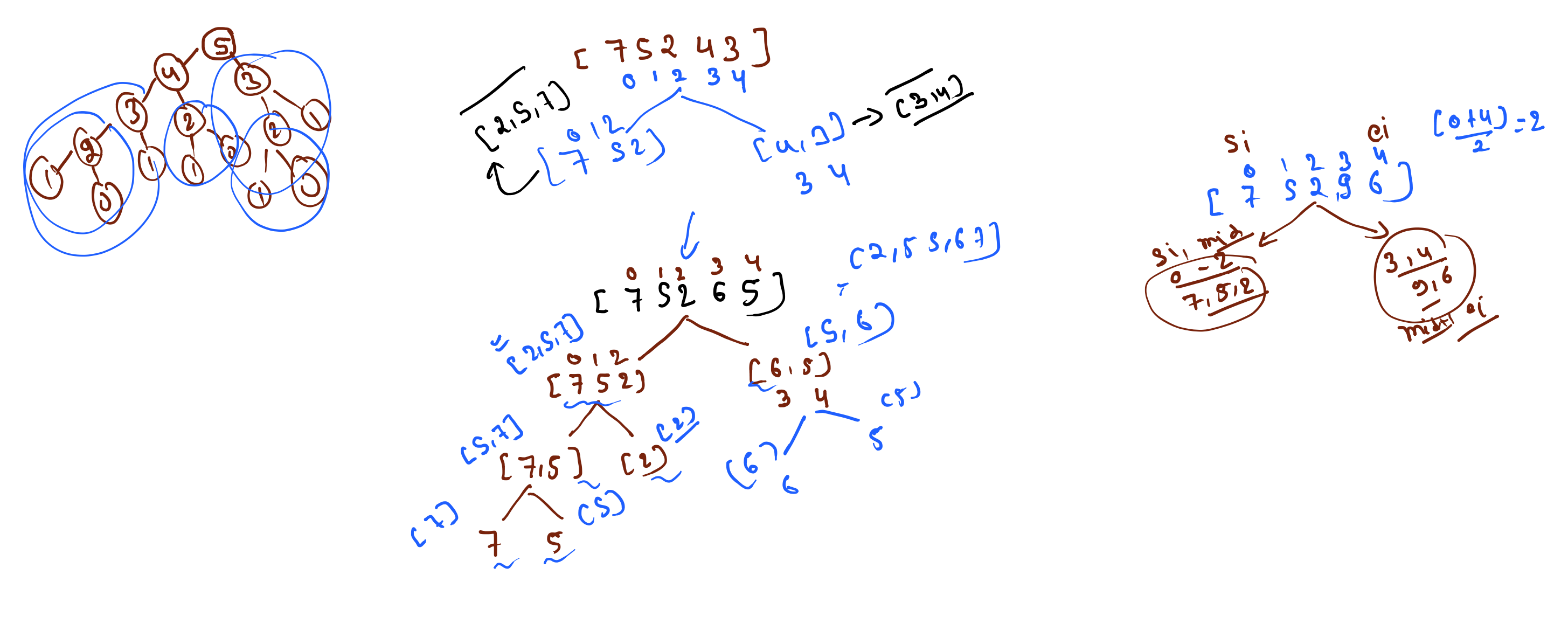

1 2 3 3 5 5 7 9 13 13 16 19
0 1 2 3 4 5 6 7 8 9 10 11
[2, 3, 5, 7, 9]
i=5
[1, 3, 5, 11, 13, 16, 19]

```
int[] arr = { 5, 7, 2, 1, 8, 3, 4 };
```

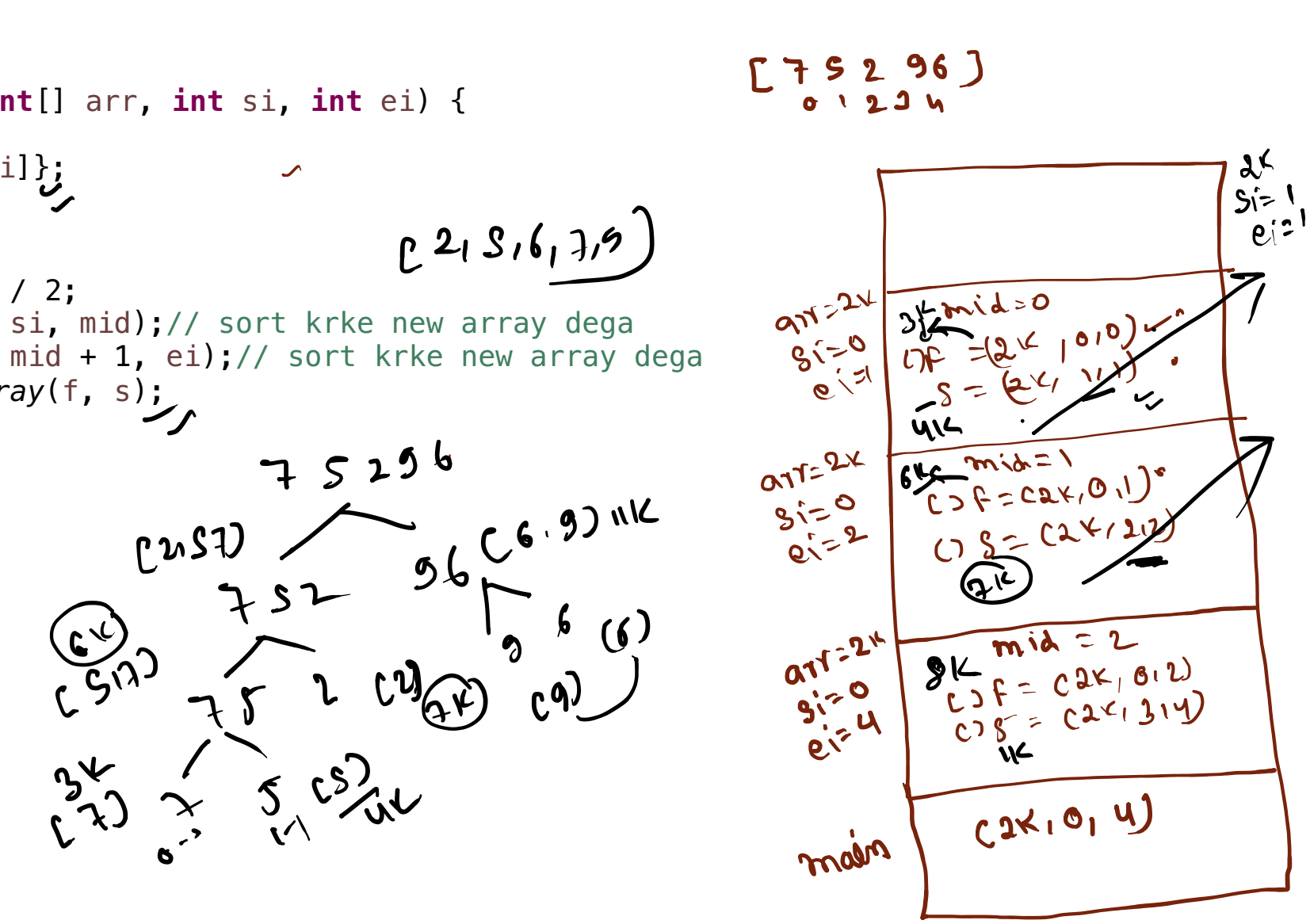

O(n)
Left
<4<
Right
1 2 3 4 5 7 8 n/n
2 1 3 4 8 5 7


ans
s/m
[5, 7, 2, 1, 8, 3, 4]
si ei
0 1 2 3 4 5 6
2 7 5 1 8 3 4
2 1 5 7 8 3 4
2 1 3 7 8 5 4
2 1 3 4 8 5 7
swap
3,6
item = arr[ei] = 6
idx = si = 0 1 2 3
for(i=si; i<ei; i++){
if(arr[i] < item){
swap(i, idx)
idx++
}
}
ei, idx
idx


D&C
Pre
Solve
Solve
Merge
5
4
3
D&C


[7 5 2 4 3]
0 1 2 3 4
[2,5,7]
[7 5 2]
[4,3] → [3,4]
[7 5 2 6 5]
[2,5,5,6,7]
[2,5,7]
[7 5 2]
[6,5]
[5,6]
[5,7]
[7,5]
[2]
[7]
[5]
[6]
[5]
si ei
[7 5 2 9 6]
(0+4)/2 = 2
si, mid
0-2
7,5,2
3,4
9,6
mid+1, ei

```
public static int[] Sort(int[] arr, int si, int ei) {
    if (si == ei) {
        int[] a = {arr[si]};
        return a;
    }
    int mid = (si + ei) / 2;
    int[] f = Sort(arr, si, mid);// sort krke new array dega
    int[] s = Sort(arr, mid + 1, ei);// sort krke new array dega
    return Merge_Two_Array(f, s);
}
```


[7 5 2 9 6]
0 1 2 3 4
[2, 5, 6, 7, 9]
7 5 2 9 6
[2,5,7]
7 5 2
9 6 [6,9] 11K
[5,7]
[7,5]
2
[2]
[9]
[6]
[7]
[5]
mid=0
mid=1
mid=2
arr=2K si=0 ei=1
arr=2K si=0 ei=2
arr=2K si=0 ei=4
(2K,0,4)
main